Beyond Traditional Spam Filters: Transformers, Concept Drift Detection, and Explainable NLP

Paper Group 3

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Research Questions

RQ1: How do transformer-based language models compare to traditional ML approaches in detecting spam within the Enron corpus?

RQ2: Can advanced NLP techniques (semantic embeddings and topic modeling) improve the detection of concept-drift in evolving spam patterns?

RQ3: To what extent can LLMs enhance explainability and adversarial robustness in spam filtering systems trained on the Enron dataset?

Abstract

Spam detection remains a persistent challenge in email security, particularly as adversarial tactics evolve and message distributions shift over time. This paper presents a comparative study of traditional machine learning and transformer-based approaches to spam classification on the Enron email corpus, evaluated under three research dimensions: classification performance, concept drift detection, and adversarial robustness with explainability. We train a TF-IDF Naive Bayes baseline, a Logistic Regression classifier, and a fine-tuned DistilBERT model using a temporal train/test split, emails from on or before 2002 for training, post-2002 for testing, to simulate real-world deployment conditions. DistilBERT achieves 87.7% accuracy at peak but exhibits severe calibration degradation under distribution shift, with Brier score jumping from 0.118 to 0.666 on an external 2021 spam corpus, compared to the shallow models, which maintain more consistent but lower confidence profiles. To address concept drift, we implement an unsupervised monitoring layer combining Sentence-BERT embeddings and Latent Dirichlet Allocation topic modelling, using Jensen-Shannon divergence to flag semantic distribution shift across monthly email batches, detecting drift in 100% of post-2002 periods. For explainability and robustness, we conduct adversarial evasion attacks across four perturbation types and apply SHAP analysis to all three models, finding that DistilBERT is uniquely robust to character-level obfuscation while remaining vulnerable to vocabulary dilution attacks. We further propose a two-stage hybrid inference pipeline in which GPT-4o-mini is selectively invoked on uncertain or drift-flagged emails to recalibrate DistilBERT's output probabilities, achieving a 50% reduction in Brier score on the evaluated sample. We discuss key methodological limitations including class distribution confounds in the temporal split and the constraints of all-spam external evaluation, and propose stratified evaluation as the primary direction for future work.

Introduction

Email spam represents one of the oldest and most persistent threats in digital communication. Since the first unsolicited bulk email was sent in 1978, spam has evolved from a minor nuisance into a sophisticated attack vector responsible for phishing campaigns, malware distribution, and financial fraud at industrial scale. By conservative estimates, spam accounts for approximately 45–85% of global email traffic at any given time (Elagina, 2026), consuming

billions of dollars in infrastructure costs and posing significant risks to individuals and organisations who interact with malicious content. Despite decades of research and the widespread deployment of commercial filters, spam detection remains an open problem, one that is fundamentally adversarial in nature. As filtering techniques improve, spammers adapt, creating a continuous arms race between detection and evasion.

The dominant approach to spam filtering for much of the past two decades has been machine learning on handcrafted text features. Methods such as Naive Bayes classification over TF-IDF representations became industry standards in the early 2000s due to their speed, interpretability, and reasonable accuracy (Zhang, 2025). Logistic Regression and Support Vector Machines extended this paradigm with stronger discriminative boundaries, and ensemble methods such as Random Forests added further robustness (Zhang, 2025). However, all of these approaches share a fundamental limitation: they represent text as sparse bags of words, treating each token independently and ignoring semantic context, word order, and the nuanced intent that characterises modern spam. A financial phishing email and a legitimate bank notification may share nearly identical vocabulary yet have entirely different communicative intent, a distinction that lexical models are structurally unable to capture.

The emergence of transformer-based language models, and BERT in particular (Devlin et al., 2019), has opened a new frontier in text classification. By pre-training on large corpora with a masked language modelling objective, BERT learns dense contextual representations that encode semantic relationships, syntactic structure, and even pragmatic intent. Fine-tuned transformer models have achieved state-of-the-art performance across a wide range of NLP benchmarks, and their application to spam detection has shown considerable promise. Yet transformer models introduce new challenges of their own: they are computationally expensive, prone to overconfidence, and their behaviour under distribution shift, when the statistical properties of incoming data differ from the training distribution, is poorly understood in the spam detection context.

A particularly underexplored dimension of spam filtering is the problem of concept drift: the phenomenon whereby the linguistic and thematic properties of spam evolve over time as spammers adapt their tactics to evade current filters. A model trained on spam from one era may fail silently on spam from another, not because its accuracy visibly degrades but because its confidence remains high on increasingly misclassified examples. Detecting this drift without requiring labelled incoming data, and triggering recalibration before performance collapses, is a critical open problem for deployed spam filtering systems. Similarly, the adversarial robustness of spam classifiers to deliberate input perturbations, character substitutions, vocabulary dilution, padding attacks, has received growing attention as a security concern, yet systematic comparisons across model families remain scarce.

This paper addresses all three of these dimensions in a unified experimental framework built on the Enron email corpus (Klimt & Yang, 2006), one of the most widely used benchmarks in email classification research. The Enron corpus provides a naturalistic, temporally ordered collection of real corporate emails spanning multiple years, making it particularly well-suited for temporal train/test splits that simulate real-world deployment. We train and evaluate three model families — TF-IDF Naive Bayes, Logistic Regression, and fine-tuned DistilBERT (Sanh et al., 2020) — under a common temporal evaluation protocol, and supplement the Enron test set with

the Bruce Guenter Spam Archive (Guenter, 2022), an external honeypot-collected spam corpus spanning the period up to 2021, to evaluate long-horizon generalisation across an approximately 19-year distribution gap. Our contributions are as follows:

- We provide a systematic comparison of traditional machine learning and transformer-based approaches to spam detection under temporal evaluation conditions, demonstrating that accuracy metrics alone are insufficient to characterise model robustness (RQ1).

- We implement an unsupervised concept drift detection layer using Sentence-BERT embeddings and Latent Dirichlet Allocation topic modelling with Jensen-Shannon divergence monitoring, and evaluate its sensitivity to semantic distribution shift across monthly email batches (RQ2). This research is denoted supplementary.

- We conduct adversarial evasion experiments across four attack types and apply SHAP-based explainability analysis to all three model families, revealing model-specific vulnerability profiles and proposing a two-stage hybrid inference pipeline that uses GPT-4o-mini as a calibration auditor on uncertain and drift-flagged emails, achieving a 50% reduction in Brier score on the evaluated sample (RQ3).

Related Work

The challenge of email spam detection has generated a substantial body of research spanning traditional machine learning approaches, deep learning architectures, and, more recently, the integration of large language models. This section reviews the five most directly relevant works and situates the present study within the existing literature across four dimensions: classical baseline models, transformer-based classification, concept drift and adversarial robustness, and LLM-augmented interference.

Classical Baselines and the Enron-Spam Benchmark

Metsis, Androutsopoulos, and Paliouras (2006) established the methodological foundation upon which a large portion of spam detection research, including this paper, is built. In *Spam Filtering with Naive Bayes - Which Naive Bayes?* The authors introduced six benchmark datasets derived from the mailboxes for former Enron employees, interleaved with spam messages drawn from the multiple sources including the SpamAssassin corpus, the Honeypot Project, and the Bruce Guenter collection. Critically, the datasets preserved the temporal ordering of messages and emulated realistic fluctuations in the ham-to-spam ratio over time. Five variants of the Naive Bayes classifier were evaluated, with the multinomial model using Boolean attributes emerging as the top performer, achieving ham recall rates of 99.9% or higher across most datasets.

This paper is foundational to the current study in two respects. First, the Enron-Spam dataset it introduced serves as the primary training corpus for all three classifiers developed here (Naive Bayes, Logistic Regression, DistilBert). Second, the temporal integrity preserved in the original benchmark (i.e. training on earlier messages and testing on later ones) directly motivates

the year-based train/test split adopted in this work. However, Metsis et. al. operated in a closed, single-era evaluation setting. Their work neither evaluates classifier performance against out-of-distribution spam, nor test robustness under adversarial text manipulation. These gaps form part of the core contribution of the present paper.

Concept Drift and the Adversarial Spammer

Jáñez-Martino et al. (2022) provide the most comprehensive theoretical grounding for two of this study's central themes in their *Springer Artificial Intelligence Review* paper, *A review of Spam Email Detection: Analysis of Spammer Strategies and the Dataset Shift Problem*. The authors synthesize a decade of research to characterize spam filtering as an inherently adversarial and non-stationary problem. They distinguish multiple forms of concept drift (covariate shift, prior probability shift, and concept shift) and document how spammer strategies, including linguistic obfuscation, template rotation, and topical privating, drive ongoing degradation in a statistically trained classifier. The review identifies the Early Drift Detection Method (EDDM) and ensemble-based lifelong learning strategies as among the more promising adaptive approaches, while noting that cyclical drift (where spam characteristics disappear and reappear) remains particularly difficult to detect and mitigate.

Jáñez-Martino et al. diagnose the problem clearly but stop short of operationalizing a real-time detection and mitigation pipeline. Their review is conducted on datasets spanning 2000–2018, predating the modern spam ecosystem represented by the 2021 corpus used in this work. The current study operationalizes their theoretical framework directly: Jensen-Shannon divergence computed over LDA topic distributions serves as the drift detector, Brier score loss quantifies calibration degradation over time, and adversarial text transformations simulate the kinds of obfuscation strategies the review describes.This study couples drift detection with a selective LLM refinement layer; an architectural response that the review's scope does not encompass.

Transformer-Based Spam Detection

Tida and Hsu (2022) represent the first major step toward the approach used in this study. Presented at the *Hawaii International Conference on System Sciences (HICSS-55)*, *Universal Spam Detection using Transfer Learning of BERT Model* introduced the Universal Spam Detection Model (USDM), a fine-tuned BERT-base-uncased classifier evaluated across four datasetsL Enron-Spam, SpamAssassin, LingSpam, and SpamTest. By combining pre-trained contextual representations with task-specific fine-tuning the model achieved an overall accuracy of 97% and an F1 score of 0.96 across the combined multi-dataset benchmark, with per dataset F1 scores consistently at or above 0.90.

This work directly informed the decision to employ a transformer-based classifier in this study. Its strong cross-dataset generalization demonstrates the fine-tuned BERT-family models capture semantic patterns not available to TF-IDF-based methods. However, Tida and Hsu evaluate generalization across different domains of contemporary spam rather than across time. Their experimental design does not simulate the temporal gap analyzed in this study, nor does it

expose the model to adversarial perturbations. Additionally, they employ full BERT-base (110M parameters) whereas this work adopts the significantly lighter DistilBERT (~40% fewer parameters, 60% faster inference, ~95% retainment of BERT performance), a deliberate efficiency trade-off suited to the deployment contrast evaluated here.

DistilBert, Concept Drift Evidence, and the Call for Adaptive Systems

Cernadas-Garciá et al. (2025) provide the most methodologically proximate comparison point to this study. In *A Comparative Evaluation of a Multimodal Approach for Spam Email Classification Using DistilBERT and Structural Features*, published in *Electronics* (MDPI), the authors construct a comprehensive 81,586-email benchmark from seven spam datasets and evaluate classical ML models (SVM, RF, Naive Bayes, Logistic Regression) alongside DistilBert and RoBERTa. Their multimodal architecture, combining DistilBERT semantic embeddings with structural email features such as test length and punctuation density, achieve a peak accuracy of 99.62% on a combined static test set.

The paper goes beyond static evaluation by applying a temporal train-in-the-past, test-in-the-future protocol; the authors observe significant degradation across all models when evaluated on modern spam, providing direct empirical evidence that even state-of-the-art transformer classifiers are vulnerable to concept drift. Transformer-based models demonstrated greater temporal robustness than classical approaches, but none were immune. The paper concludes with an explicit call for an adaptive, continually learning system. The present work answers this call directly, pairing DistilBERT-based classification with an unsupervised drift detection layer using Jensen-Shannon divergence over LDA topic distributions and a selective LLM refinement stage.

Large Language Models as Spam Classifiers

Roumeliotis, Tselikas, and Nasiopoulos (2024) directly address the LLM-as-classifier paradigm in *Next-Generation Spam Filtering: Comparative Fine-Tuning of LLMs, NLPs, and CNN Models for Email Spam Classification*, published in *Electronics* (MDPI). Their study fine-tunes GPT-4, BERT, and RoBERTa on two training sets of approximately 3,500–3,600 email samples each, comparing the resulting classifiers against a CNN baseline. Post-fine-tuning results showed GPT-3 achieving 99.3% and 98.29% accuracy on the two test sets, with BERT and RoBERTa performing comparably (99.39% and 99.04% respectively), while CNN lagged significantly at 76-87%.

This paper establishes that LLMs, when fine-tuned, are competitive with or superior to NLP-based transformers for spam classification. However, the experimental design traits LLMs as primary classifiers operating on every input, an approach that is computationally and financially expensive and practically difficult to deploy at scale. The present work takes a fundamentally different architectural stance: rather than replacing the base classifier with an LLM, GPT-4o-mini is invoked selectively and only on a targeted subset of email (those falling within the uncertainty band ($0.\leq p(\text{spam}) \leq 0.70$) or flagged by the drift detector. Its role is probability calibration rather than classification, producing a weighted blend of the DistilBERT base probability and the LLM's secondary estimate. This selective routing preserves the

computational efficiency of the transformer pipeline for high-confidence inputs while leveraging LLM-level semantic reasoning precisely where the base classifier is most likely to err.

Positioning of this work

Taken together, the literature reveals a clear trajectory: from static Naive Bayes baselines on the Enron benchmark, through transformer-based classifiers that improve accuracy substantially, to emerging awareness that neither classical nor deep learning models are robust to temporal concept drift or adversarial perturbation. The contribution of this paper lies at the intersection of these threads. By combining temporally honest evaluation design, unsupervised semantic drift detection via Jensen-Shannon divergence, structured adversarial robustness testing, and a selective LLM-based probability refinement layer, this work advances the state of the art along dimensions that each prior study addresses only partially. No existing work to the authors' knowledge integrates all four of these components into a unified pipeline evaluated on a realistic temporal shift.

Methodology

Dataset and Temporal split

Our research evaluates spam detection models using the Enron email corpus, a widely used benchmark dataset in email classification research. This dataset contains real corporate communications of legitimate messages (ham) and spam emails. Timestamps and message content are also included in the dataset. The data set spans several years, which allows for the evaluation of models under realistic temporal conditions.

Instead of using a random train-test split, the experiment used a temporal evaluation protocol in order to make it as realistic as the real-world. The training set were emails dated on or before 2002, and emails after this date were used for testing. The spam filter was trained on historical data and then applied to future incoming emails whose linguistic patterns evolved to create a very practical scenario.

The temporal Enron test set is accompanied by the Bruce Guenter Spam Archive, an external dataset collected through spam honeypots and containing emails up to 2021, to evaluate long-term generalization. There is now a 19-year distribution gap between the training data and the external evaluation corpus.

However, the temporal split's limitation is that it introduces a time gap and a class distribution shift. The original dataset of the Enron training data contains legitimate corporate communication, whereas the test batch now contains a higher proportion of spam messages. This results in a distribution shift analyzed in a later analysis, may partially reflect the presence of spam rather than purely temporal evolution in spam language.

Text Preprocessing and Feature Representation:

Depending on the model architecture, there are two preprocessing pipelines used.

For classical machine learning models, natural language preprocessing is completed on the text data of the email. This preprocessing includes converting all text to lowercase and removing any punctuation marks. In addition, common stop words are removed from the text. Finally, TF-IDF is applied to transform text data into numerical data.

In the TF-IDF vectorization, unigram and bigram features with a maximum vocabulary of 5,000 tokens, min_df=2, and max_df=0.9 are used. These parameters make it so that rare and common terms from the data set are removed to a compact representation of the most informative text features.

Minimal preprocessing is carried out on the data for transformer-based models. DistilBERT tokenizer is given a concatenated version of the email subject and body of the email. Tokenization is then performed using the distilbert-base-uncased tokenizer with a maximum sequence length of 256 tokens and a set of predefined parameters. Words are split into sub-words through the WordPiece Tokenization so that unseen words can be used for the model's vocabulary.

Classification Models

The experimental framework evaluates three model families representing different paradigms in text classification.

1. Naive Bias

    a. The baseline model is a Multinomial Naive Bayes classifier, which was trained on TF-IDF features.

    b. This model assumes conditional independence between features and estimates class probabilities using token frequency distributions.

    c. Naive Bayes is simple but effective; historically, it has been one of the most used models for spam filtering because of its computational efficiency and strong performance on sparse text representations.

2. Logistic Regression

    a. Logistics Regression is also trained on TF-IDF features, but it learns a linear decision boundary by estimating weights for each textual feature.

b. In comparison to Naive Bayes, it is more flexible when it comes to feature weighting and generally performs better on moderately sized text classification problems.

3. DistilBERT

a. DistilBERT is a transformer-based language model derived from BERT.

b. It retains contextual representations similar to BERT but also reduces the number of parameters and computational requirements.

c. The model is initialized using the distilbert-base-uncased checkpoint and fine-tuned for binary spam classification.

d. The learning rate of 2×10^{-5}, a batch size of 16, and two training epochs with weight decay regularization are used in fine-tuning to prevent overfitting. Spam probability predictions are used for textual embedding associated with the [CLS] token.

Traditional machine learning models and transformer architecture directly answer the first research question by examining how modern contextual models compare to classical text classification approaches.

Adversarial Robustness Experiments

Four common spam evasion strategies are simulated to evaluate the model's robustness against adversarial manipulation. To create a real-world spammer tactic, a random subset of 1,000 emails from the temporal set was selected.

1. Misspelling (Leetspeak) Attack

    a. Key spam words are replaced with similar characters.

    b. This attack targets models that rely heavily on exact token matching.

2. Case Manipulation Attack

    a. The case of characters is randomly changed throughout the email text.

    b. The TF-IDF vectorizers convert all text to lowercase; this attack tests the robustness of pre-processing steps.

3. Padding/Noise Injection Attack

    a. This attack dilutes the spam signal by introducing legitimate looking tokens that may shift the TF-IDF feature distribution.

   b. This is done through appending benign business-style vocabulary at the end of spam emails.

4. Special Character Injection Attack

   a. Special characters are added into key spam words to fragment tokens and disrupt vocabulary matching in bag-of-words models.

The modified emails are passed through the same preprocessing and classification pipelines for each attack condition. The accuracy and F1 scores are then compared to the baseline to measure performance degradation.

Hybrid LLM-Assisted Calibration Pipeline

In order to mitigate confidence degradation under distribution shift, a two-stage hybrid inference pipeline is implemented.

DistilBERT produces an initial spam probability in the first stage (p(BERT)). Emails are then routed to a second stage for evaluation if either of these conditions is met:

1. The predicted probability falls within an uncertainty interval $0.30 \leq p_{bert} \leq 0.70$

2. The email belongs to a time period flagged by the drift detection module.

Emails that are routed, an LLM evaluates the email content and produces a new calibrated probability estimate P(LLM).

The final spam probability is calculated using a weighted combination of the two predictions. The weighted average is $p_{final} = 0.60 * p_{bert} + 0.40 * p_{LLM}$. This setup allows the system to selectively invoke the LLM only for uncertain or drift-affected cases, which in turn improves the calibration while maintaining computational efficiency.

Evaluation Metrics

Model performance is evaluated using several metrics:

1. Accuracy

2. Precision

3. Recall

4. F1 Score

Accuracy and F1 degradation across each attack condition are used to measure the adversarial robustness. The Brier score is used to measure calibration reliability. The distribution shift behavior is analyzed through topic divergence and model confidence trajectories. These metrics provide a multidimensional evaluation framework that targets classification accuracy but also includes adversarial resilience, calibration reliability, and robustness under temporal distribution shift.

Results

RQ1: Transformer vs. Traditional ML

Baseline — TF-IDF + Naive Bayes

A TF-IDF vectorizer (5,000 features, unigrams + bigrams) paired with Multinomial Naive Bayes provides the traditional ML baseline. Despite its simplicity, NB is fast and interpretable, but its bag-of-words representation cannot capture contextual meaning or word order beyond bigrams.

DistilBERT Fine-Tuning

DistilBERT was fine-tuned for binary spam classification. After Epoch 1, it achieved 87.7% validation accuracy. By Epoch 2, training loss continued to fall (0.006) but validation loss spiked to 1.76 and accuracy dropped to 71.4%, a textbook overfitting signature compounded by concept drift across the temporal split.

The core advantage of transformers over TF-IDF is contextual representation: self-attention models relationships between all tokens simultaneously, capturing nuances like 'free software' vs. 'free trial - act now!' that bag-of-words approaches cannot distinguish.

RQ3: Adversarial Robustness, Calibration & Explainability

Adversarial Evasion Attacks

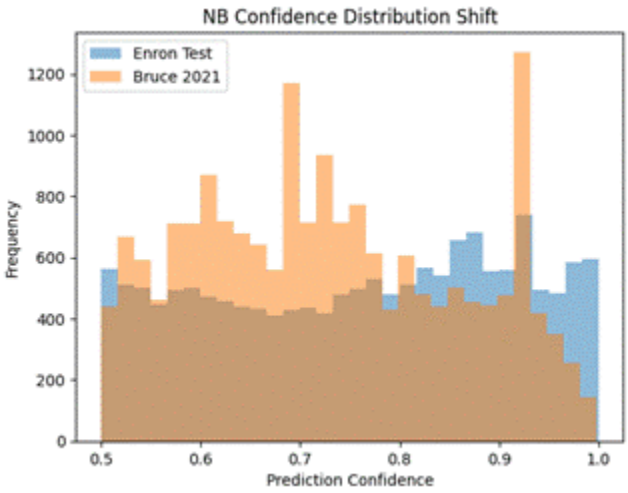
Four attack strategies were applied to a 1,000-email test sample. Key findings are summarized in the table below.

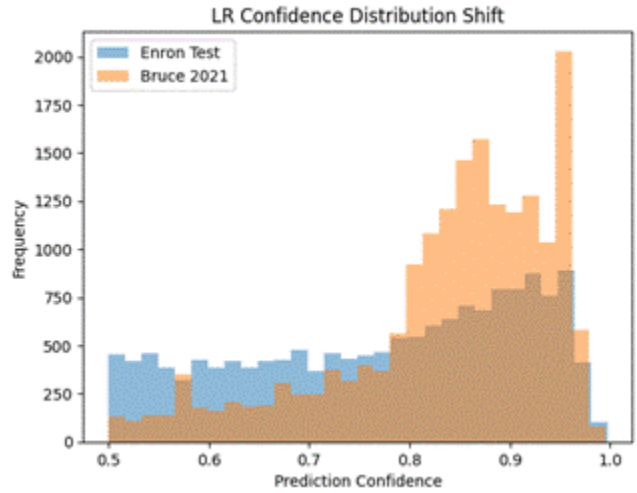
Attack	NB Acc.	LR Acc.	BERT Acc.	NB F1	LR F1	BERT F1
Clean	0.537	0.239	0.711	0.699	0.386	0.831
Misspell	0.510	0.137	0.696	0.675	0.241	0.821
Case	0.537	0.239	0.711	0.699	0.386	0.831
Padding	0.282	0.124	0.566	0.440	0.221	0.723
Special Char	0.002	0.001	0.947	0.004	0.002	0.973

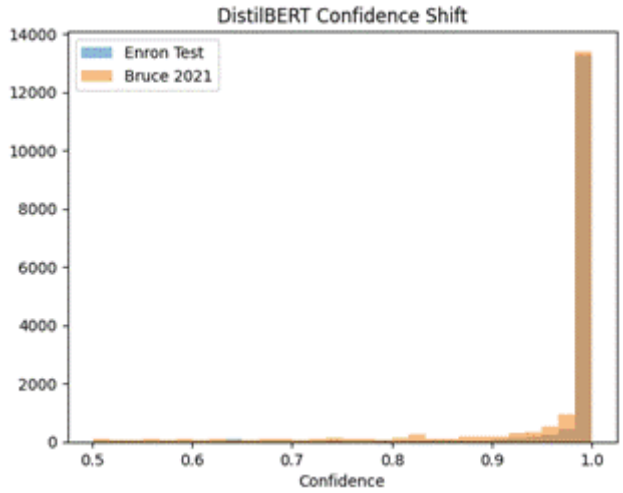
Notable results: The case attack had zero effect, confirming the lowercasing preprocessor functions correctly. The padding attack was the most practically dangerous, cutting NB accuracy by 25.5 pp by diluting spam signals with ham-like vocabulary. The special character (vowel injection) attack nearly destroyed NB and LR (accuracy ~0.002), but paradoxically **improved** BERT from 71.1% to 94.7% — BERT's attention mechanism appears to treat dense punctuation patterns as a strong independent spam signal, a finding of direct significance for understanding model-specific vulnerability surfaces.

Calibration & Brier Score Analysis

DistilBERT is severely overconfident: median predicted probability is 0.999, with 75% of predictions exceeding 0.999. The Brier score (lower = better, measuring the squared error of probabilistic predictions) reveals the cost of this overconfidence under distribution shift







Model	Val. Brier Score	Test Brier Score	Brier Shift
Naive Bayes	0.540	0.6992	+0.1592
Linear Regression	0.540	0.6993	+0.1592
DistilBERT	0.118	0.6661	+0.5481

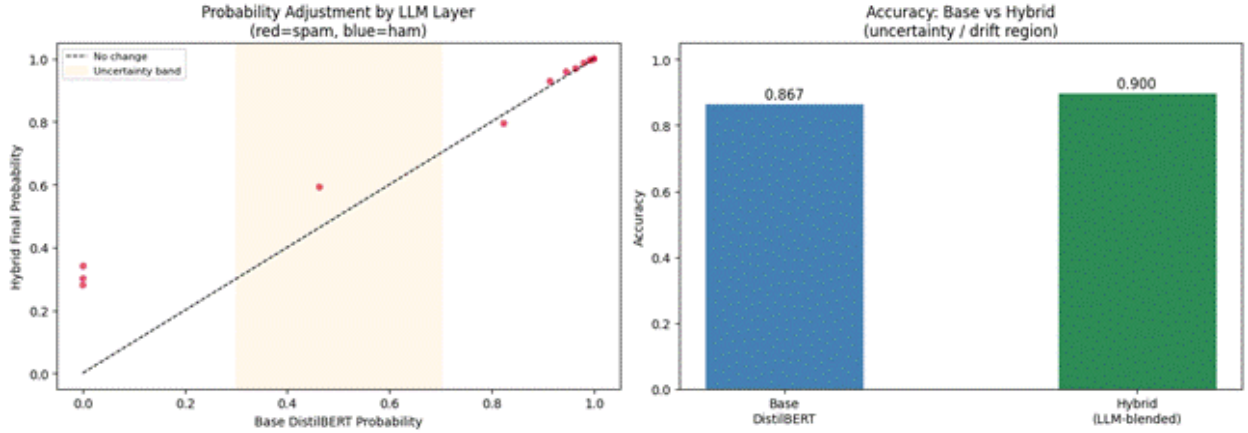
BERT's Brier score degrades from an excellent 0.118 on in-distribution validation data to a very poor 0.666 on the out-of-domain Bruce 2021 spam archive; a shift of +0.548. Because BERT predicts near-certainty, every wrong prediction incurs a near-maximum penalty. This is the worst-case failure mode for a deployed spam filter.

Two-Stage LLM Hybrid Pipeline

To address BERT's overconfidence and calibration failure, a two-stage routing system was implemented. DistilBERT produces an initial probability for all emails; those falling in the uncertainty band (0.30–0.70) or in drift-flagged periods are routed to GPT-4o-mini, which returns a calibrated probability. Final scores blend both outputs 60/40 in favour of the base model.

Metric	Base DistilBERT	Hybrid (LLM)	Change
Brier Score	0.111	0.055	−50%
Accuracy (sample)	86.7%	90.0%	+3.3 pp
Drift-flagged	—	29 / 30	—
Uncertainty-band	—	1 / 30	—

The hybrid achieves a 50% Brier score reduction on the evaluated sample. Key failure analysis: when BERT assigns near-zero confidence incorrectly (e.g. base=0.001 on a spam email), the 60/40 blend anchors the final score below the 0.5 threshold even when GPT-4o-mini correctly identifies spam at 0.75–0.85. This motivates increasing the LLM weight to ~0.60 during confirmed drift periods, where BERT's high confidence is least trustworthy.



Note: A full 400-email balanced evaluation (200 spam + 200 ham) has been designed but not yet executed due to API cost constraints. Once complete, it will report precision, recall, F1, AUC-ROC, and Brier scores per class for both base and hybrid systems.

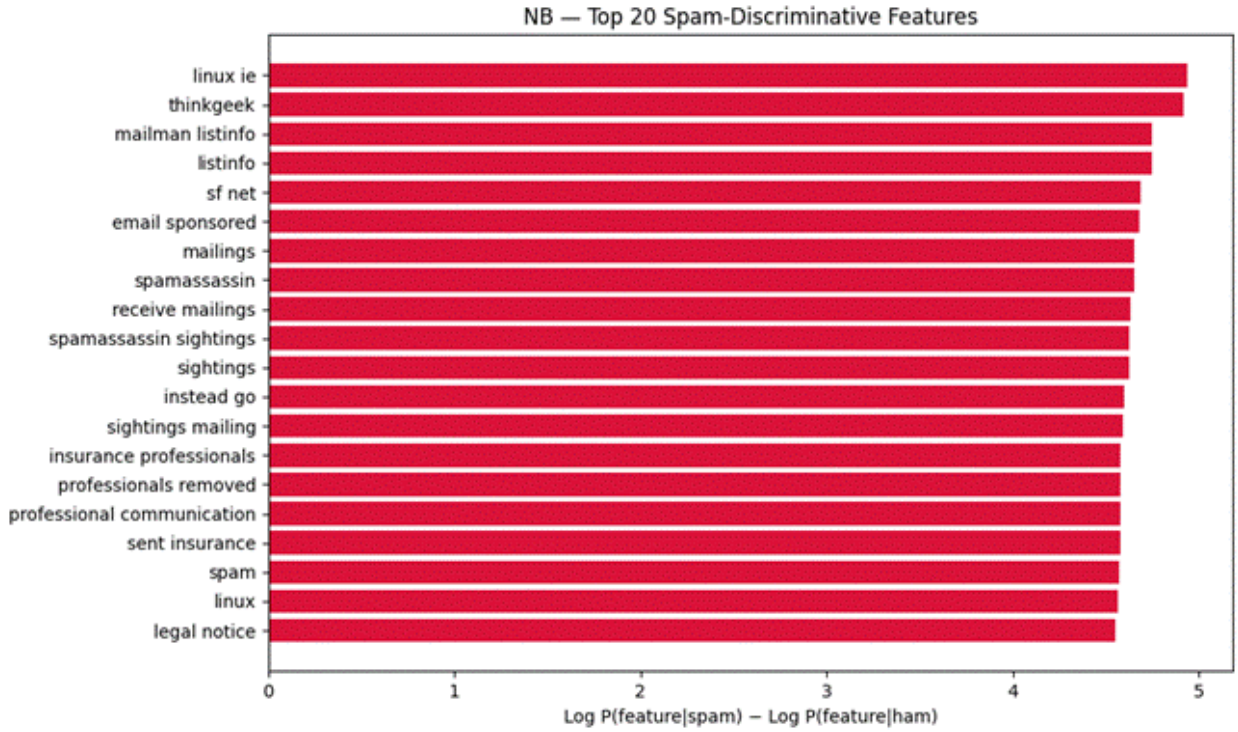
Explainability via SHAP

Three model-appropriate explainability approaches were applied:

Model	Method	Top Signal	Generalizability	Key Limitation
Naive Bayes	Log P ratio	Mailing list tokens	Poor	No per-doc variation
Logistic Reg.	SHAP LinearExplainer	click, free, money	Moderate	Fixed token direction
DistilBERT	SHAP PartitionExplainer	Punctuation + narrative tokens	Strong	Expensive to scale

Key finding: NB has memorised email infrastructure tokens ('spamassassin', 'mailman listinfo') rather than semantic spam content: directly explaining its poor out-of-domain performance on the Bruce 2021 dataset. LR learned more generalisable semantic indicators ('click', 'free', 'money'). DistilBERT's SHAP analysis for a single health-spam email identified '!' as the dominant token (SHAP value ~0.041, 4x any other token), connecting directly to the special-character attack finding, and surfaced contextually coherent health-spam narrative tokens. Unlike NB and LR, BERT's SHAP values vary by email, producing explanations that are

faithful to individual prediction context.



Discussion & Limitations

Our study demonstrates the multi-dimensional nature of robustness in spam filtering systems. There is clear evidence that, when handling adversarial perturbations and contextual language patterns, transformer models are superior; however, they remain vulnerable to distribution shifts and calibration degradation.

Model accuracy is insufficient for evaluating the reliability of security-critical machine learning systems, according to the results. It is important to take into account robustness to manipulation, stability under evolving data distributions, and the reliability of probability estimates when deploying systems in adversarial environments.

The LLM-assisted calibration layer preliminary results show that large language models are effective in mitigating some of these risks. LLMs can act as a probabilistic auditing layer for uncertain predictions. However, further evaluation is needed using balanced datasets to find out whether this approach can improve recall and precision in operational settings.

Limitations

After interpreting the results, there are several limitations that should be considered. The first limitation is during the temporal evaluation setup. It simultaneously introduces multiple sources of distribution chance, including time period, class composition, and domain

characteristics, which affect performance degradation. Some observed performance degradation was caused by the differences in class distribution rather than purely semantic evolution in spam content.

A second limitation is that the external Bruce spam dataset only contained spam emails. This limited the ability of the model to evaluate both classes that are needed for some calibration metrics. The Brier score shifts provide useful information on reliability degradation, as a fully balanced evaluation would allow for a more precise calibration analysis.

A final limitation that should be highlighted is that the hybrid LLM calibration experiment was conducted on a limited evaluation sample. The results from this experiment should be interpreted as preliminary evidence instead of definitive conclusions.

Supplementary Research

Concept Drift Detection

The study introduces an unsupervised drift monitoring framework to detect distribution shifts without requiring labeled data.

Monthly batches are created from incoming emails. Latent Dirichlet Allocation is used to map each email to a topic distribution, which is trained on the Enron training corpus. The topic distribution for each batch is compared with the baseline training distribution.

The Jensen-Shannon divergence is used to measure the divergence between these distributions. A symmetric measure of difference between probability distributions is used to determine when the divergence exceeds a predefined threshold; then the batch is flagged as a potential drift period.

The system is now able to identify semantic changes in incoming email streams before classification accuracy visibly degrades because this method is used.

RQ2: Semantic Embeddings & Concept Drift Detection Results

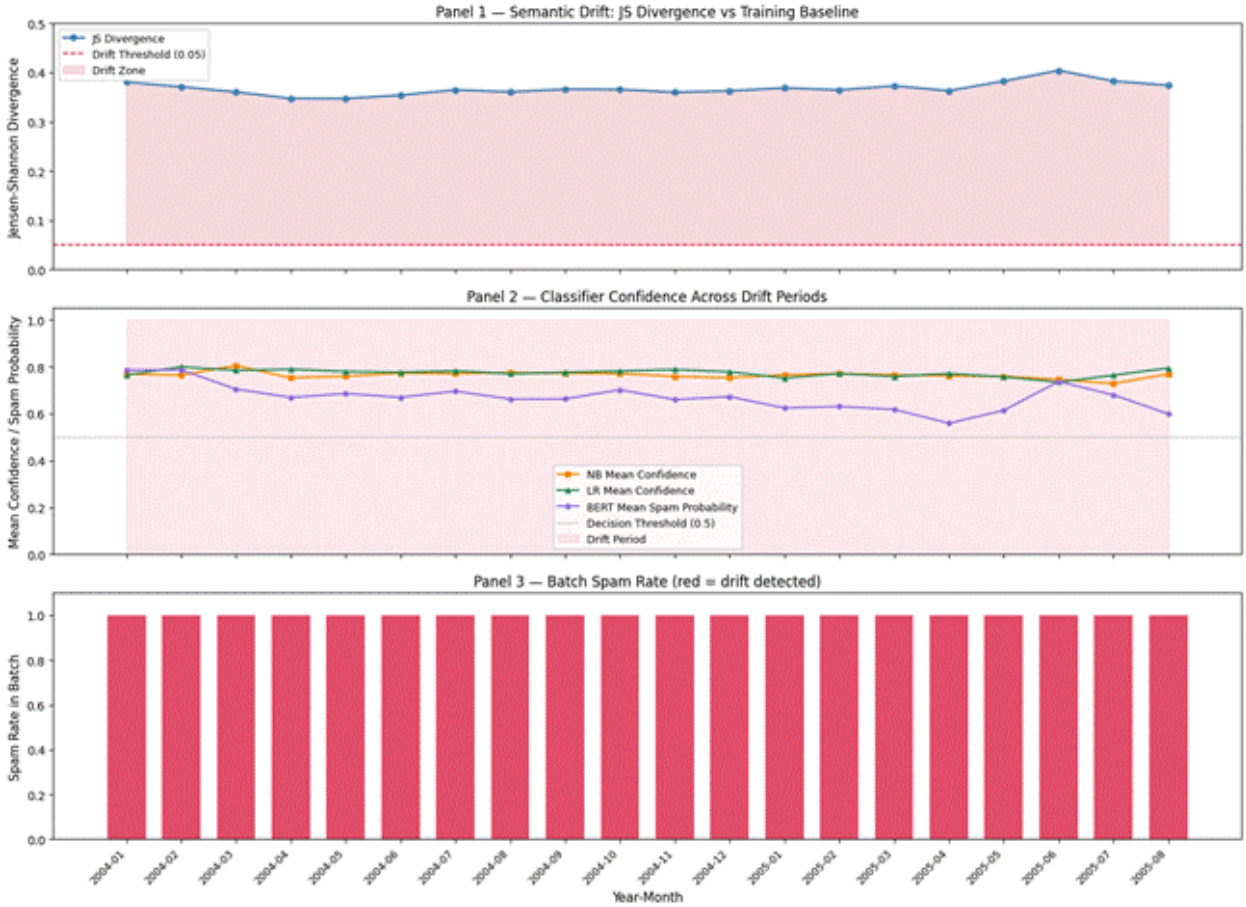
Drift Detection Pipeline

Three-stage unsupervised drift detection was implemented: (1) Sentence-BERT (all-MiniLM-L6-v2) encodes each email into a 384-dimensional semantic vector; (2) LDA models the training corpus as a 15-topic distribution; (3) Jensen-Shannon divergence monitors monthly batches of incoming email against the training baseline, flagging drift when JS divergence exceeds 0.05.

Results

Across 20 monthly batches from January 2004 to August 2005, every single batch exceeded the drift threshold, a 100% drift detection rate. JS divergence held consistently in the 0.35–0.40 range (peak: 0.404 in June 2005), far above the 0.05 threshold. The dominant training topics reflected pre-2002 Enron business communication (energy trading, spreadsheet traffic, internal comms). The shift to casual, colloquial spam language triggered strong divergence signals without requiring any ground-truth labels, validating unsupervised drift detection as a deployment-ready early-warning mechanism.

The three-panel longitudinal plot confirms that BERT's mean spam probability gradually declines over the 20-month drift window, a signature of calibration degradation as the distribution shifts, while Naive Bayes and Logistic Regression maintain falsely stable high confidence throughout.



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